

- 4 (a) Meiosis is an important process that contributes to genetic variation in a population.

Fig. 4.1 shows a pair of homologous chromosomes during prophase I of meiosis.

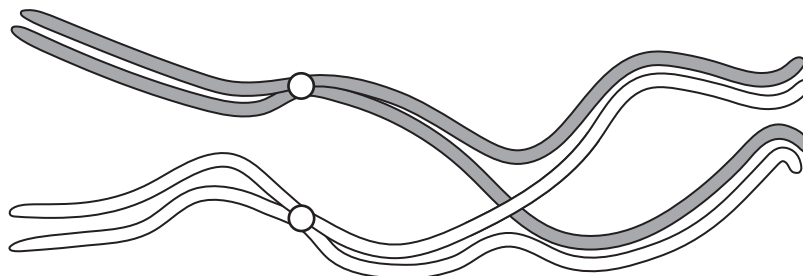


Fig. 4.1

- (i) On Fig. 4.1, use the letter **C** with a label line to identify where crossing over occurs. [1]
- (ii) Describe how crossing over produces genetic variation in a population.

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..... [4]

- (b) Wing pattern in the butterfly species *Heliconius melpomene* is controlled by genes on autosomal chromosomes.

The gene for banding pattern on the upper wing has two alleles:

- a dominant allele coding for a full band
- a recessive allele coding for a broken band.

The gene for ray pattern on the lower wing has two alleles:

- a dominant allele coding for rays
- a recessive allele coding for no rays.

Scientists crossed a butterfly that was homozygous dominant for both genes with a butterfly that was homozygous recessive for both genes. The scientists wanted to check whether the phenotypic ratio for offspring in the F₂ generation agreed with the expected phenotypic ratio of 9 : 3 : 3 : 1.

The results of this genetic cross are shown in Fig. 4.2.

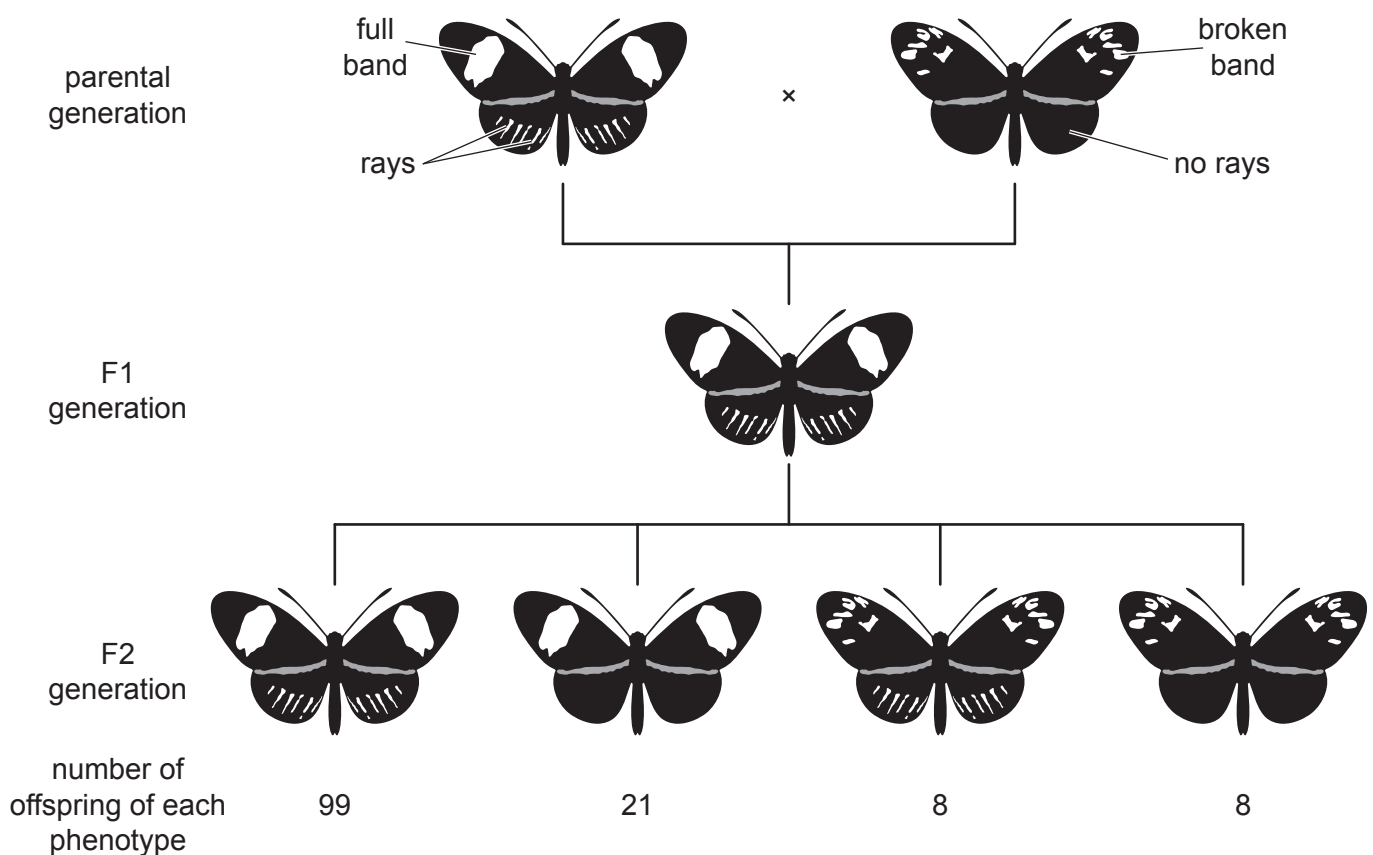


Fig. 4.2

- (i) Explain the term F1 generation.

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- (ii) The scientists used the chi-squared (χ^2) test to compare their data to the expected phenotypic ratio of 9 : 3 : 3 : 1. The formula for the chi-squared test is shown in Fig. 4.3.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

key to symbols

O = observed value

E = expected value

Fig. 4.3

Complete Table 4.1 and use the chi-squared formula in Fig. 4.3 to calculate the χ^2 value for these data.

Table 4.1

phenotype	expected ratio	O	E	$O - E$	$(O - E)^2$	$\frac{(O - E)^2}{E}$
full band and rays	9	99	76.5	22.5	506.25	6.62
full band and no rays	3	21	25.5	-4.5	20.25	0.79
broken band and rays	3	8				
broken band and no rays	1	8	8.5	-0.5	0.25	0.03

$\chi^2 =$ [3]

- (iii) The critical value at the 0.05 probability level and three degrees of freedom is 7.81.

Using the result of your calculation in (b)(ii), explain whether the results of the study agree with the expected ratio of phenotypes for the F2 generation.

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- (iv) Suggest **two** reasons why phenotypic ratios in the F2 generation do **not** always match the expected ratios.

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[Total: 15]